Acoustic Biodiversity Mapping

A global location-level biodiversity metric from Xeno-canto recordings

Generated 2026-09-08

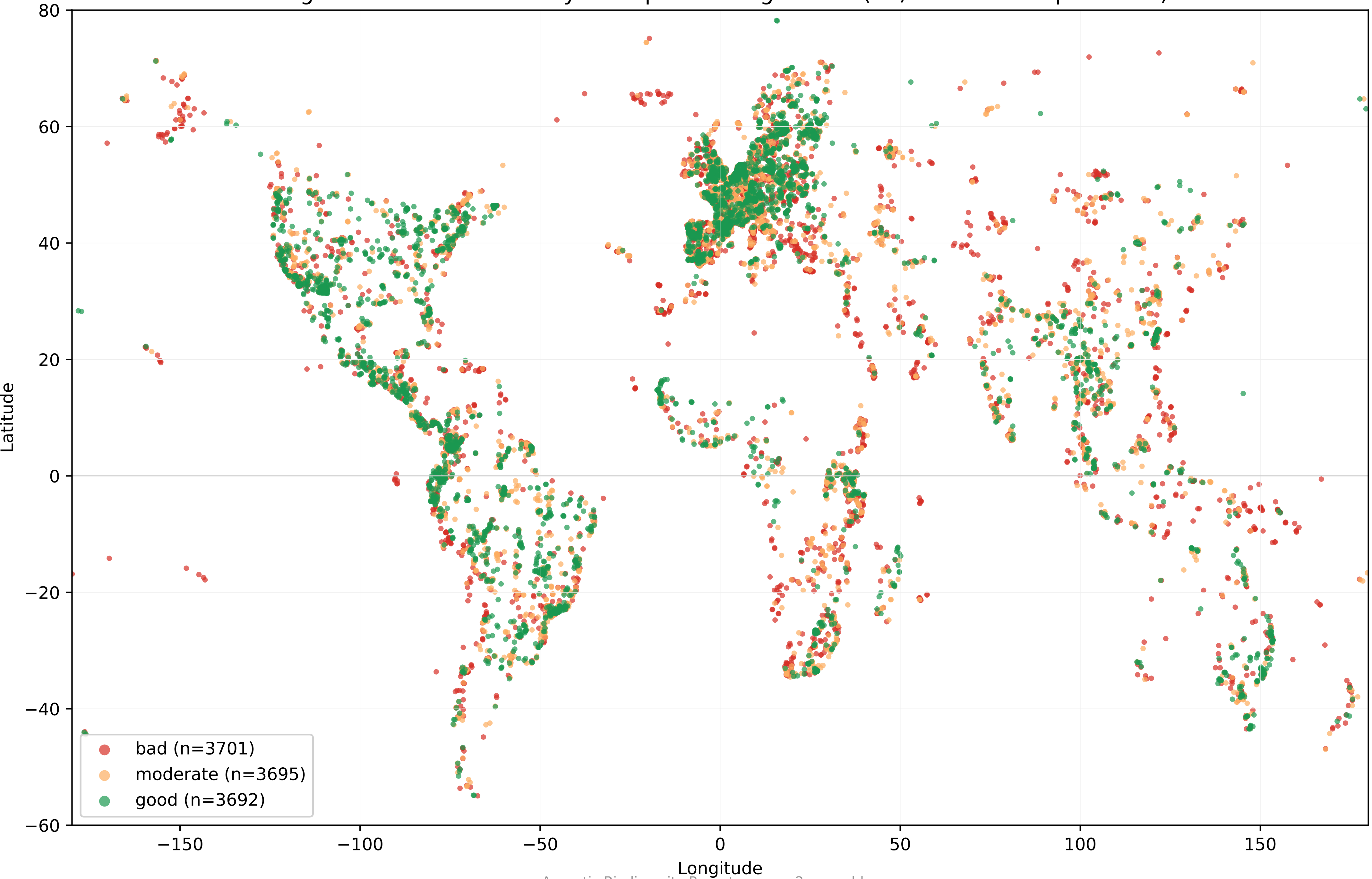
OBJECTIVE
Derive a single metric that classifies a geographic location as biodiversity 'good' or 'bad', using a global archive of wildlife sound recordings and their metadata.

DATA
- 759,767 recordings across 5 continents (Xeno-canto), each with 42 acoustic indices computed by a SLURM batch pipeline (compute_indice.py over Butterworth-filtered audio).
- Metadata flattened from 766,747 API records (species, coordinates, quality, device).
- Acoustic scores joined to metadata on recording id (>99.99% match).

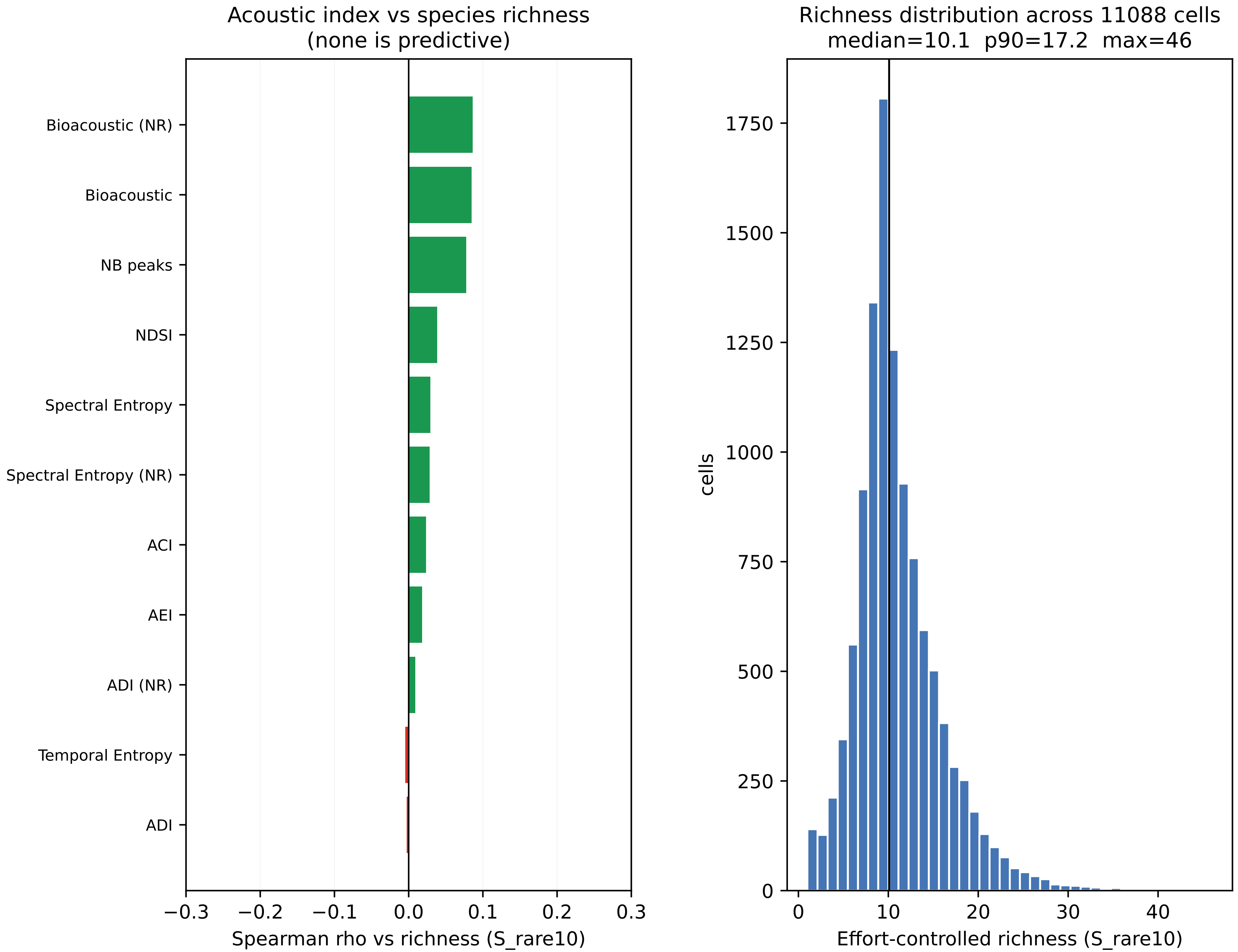
METHOD
1. Aggregate all recordings into 0.1-degree (~11 km) grid cells -> 40,914 cells.
2. Per cell, compute species richness from recorded species (genus+species and the 'also' co-occurring-species field), corrected for sampling effort by Hurlbert rarefaction to 10 recordings (S_rare10).
3. Test whether the acoustic indices predict richness (validation against ground truth).
4. Label each well-sampled cell good / moderate / bad relative to its 10-degree latitude band (region-relative tertiles).

HEADLINE RESULT
- 11,088 cells had enough recordings (>=10) to score confidently.
- The acoustic indices did NOT predict species richness (best |Spearman rho| ~ 0.09). The defensible biodiversity metric is therefore effort-controlled species richness, not any acoustic index -- the indices are unreliable on targeted (non-soundscape) recordings that dominate the archive.
- Final labels: 3692 good, 3695 moderate, 3701 bad.

Region-relative biodiversity label per 0.1-degree cell (11,088 well-sampled cells)



Validation: does the audio carry the biodiversity signal?



The Metric & Results

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METRIC DEFINITION
  location           = 0.1 deg x 0.1 deg grid cell (~11 km)
  biodiversity value = S_rare10, the expected number of distinct recorded
                       species in 10 recordings (Hurlbert sample-based
                       rarefaction; controls for recording effort)
  label              = tertile of S_rare10 WITHIN the cell's 10-deg latitude
                       band:  good = top third, bad = bottom third
  scope              = cells with >= 10 recordings (11,088 of 40,914 cells)

GLOBAL VALUE RANGE (S_rare10)
  min 1.0   p25 8.2   median 10.1   p75 13.3   p90 17.2   max 46
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Median richness by continent (well-sampled cells)

Continent	Median richness	Scored cells
america	10.9	3415
europe	10.2	4643
australia	9.9	404
africa	9.7	1003
asia	9.3	1623

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KEY LIMITATIONS (must accompany any use of this metric)
  1. Acoustic indices are non-predictive here. ACI/ADI/NDSI etc. were built for
     passive soundscape monitoring; 69% of archive clips are single-target
     recordings (median 24 s), so per-clip indices do not reflect site diversity.
  2. 'Richness' = RECORDED species, not true species. It reflects recordist effort
     and interest. Rarefaction controls sample SIZE but not observer bias.
  3. Weak latitude gradient. Effort-controlled richness is nearly flat across
     latitude (thresholds ~10-13 everywhere) -- evidence the signal is
     effort-driven, not the true tropical biodiversity peak. Region-relative
     labelling is used so scores mean 'rich for its region'.
  4. Coverage. Only 27% of cells are confidently scored; the rest are
     'insufficient_data' (< 10 recordings).
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OUTPUT FILES
  grid_cells.csv                  - per-cell features + richness (all 40,914 cells)
  grid_cells_labeled_regional.csv - region-relative good/moderate/bad labels
  biodiversity_map.csv            - lat, lon, label, code -> ready for GIS/folium
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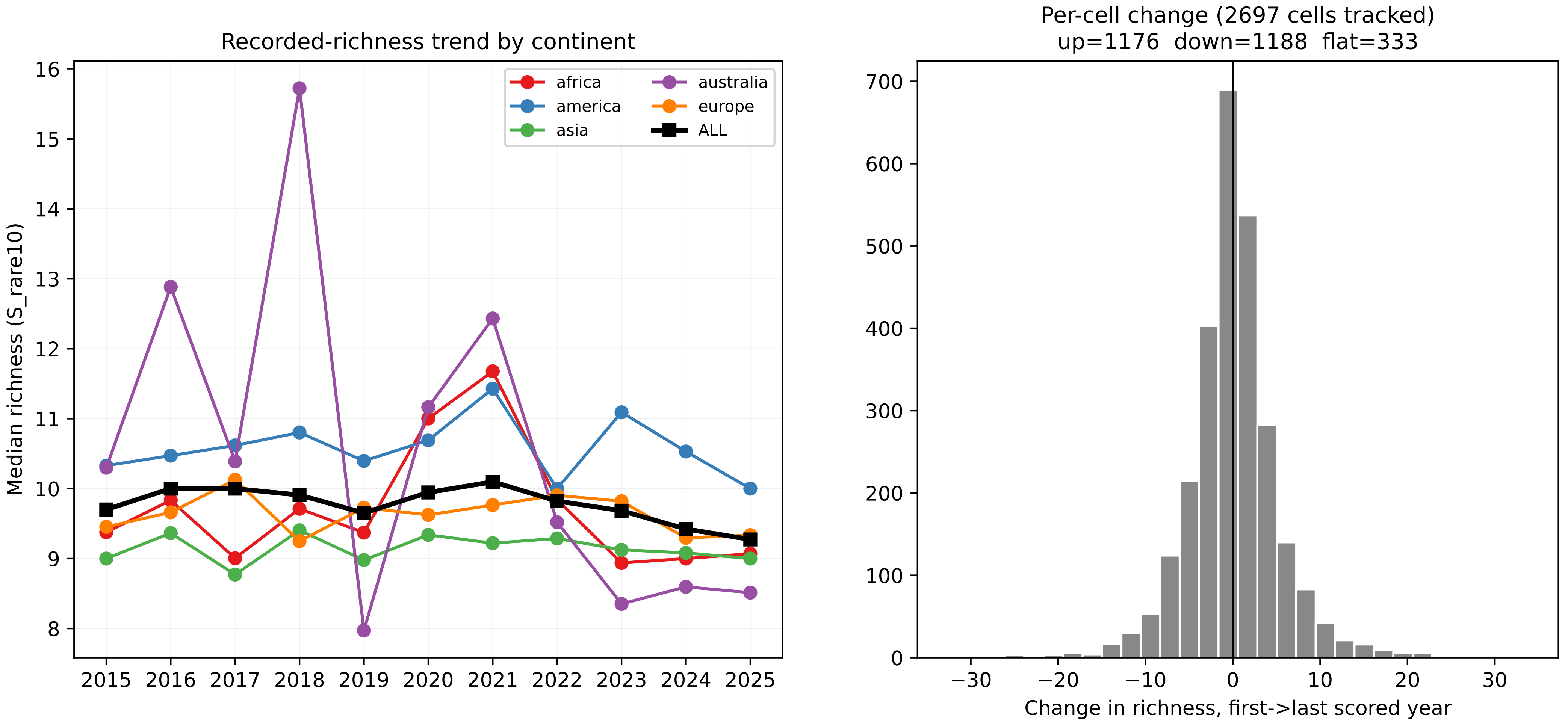
Year-by-year (2015-2025): temporal slices of the metric

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COVERAGE: 14,487 scored cell-years; 2697 cells scored in >=2 years, 10 in all 11.

CAUTION: even an 11-year window cannot show real biodiversity change. These movements reflect
WHICH cells were recorded and by WHOM each year (effort + observer turnover), not
ecological gain or loss. The global change (9.7 -> 9.3) tracks recording effort, not nature.
Use year slices to study sampling coverage over time -- not as a biodiversity time series.

OUTPUTS: grid_cells_yearly.csv (per cell-year),  cell_change.csv (per tracked cell).
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Ten-Year Findings (2015-2025)

What the decade-scale data adds to the analysis

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1.  MORE DATA MADE THE ACOUSTIC INDICES LOOK WORSE, NOT BETTER.
    Expanding from the 2023-2025 subset to the full 2015-2025 archive
    (759,767 recordings, 11,088 scored cells) shrank the best index
    correlation from |rho| ~ 0.16 to ~ 0.09. A real signal sharpens
    with more data; this faded toward zero -- the strongest evidence yet
    that the indices carry no site-biodiversity signal in this archive.

2.  A DECADE OF EFFORT-CONTROLLED RICHNESS IS ESSENTIALLY FLAT.
    Global median S_rare10 stays within 9.3-10.1 across all 11 years
    (2015->2025: 9.7 -> 9.3). This reframes the metric as a
    sampling-effort measure, not ecology. The only structure is a mild
    recent dip that tracks the incompleteness of recent uploads.

3.  TEMPORAL CHANGE IS A COIN-FLIP (RANDOM-WALK SIGNATURE).
    Of 2,697 cells tracked across >=2 years, 1,176 rose and 1,188 fell
    (flat=333). That near-perfect symmetry over a decade is the
    fingerprint of observer turnover and noise, not directional change.

4.  SPATIALLY DECADE-RICH, BUT TEMPORALLY STILL STARVED.
    Coverage grew to 40,914 cells / 11,088 scored -- a much denser map --
    yet only 10 cells were recorded in all 11 years (2,697 in >=2).
    The biodiversity MAP is far stronger; a true TIME SERIES remains
    impossible from this data.
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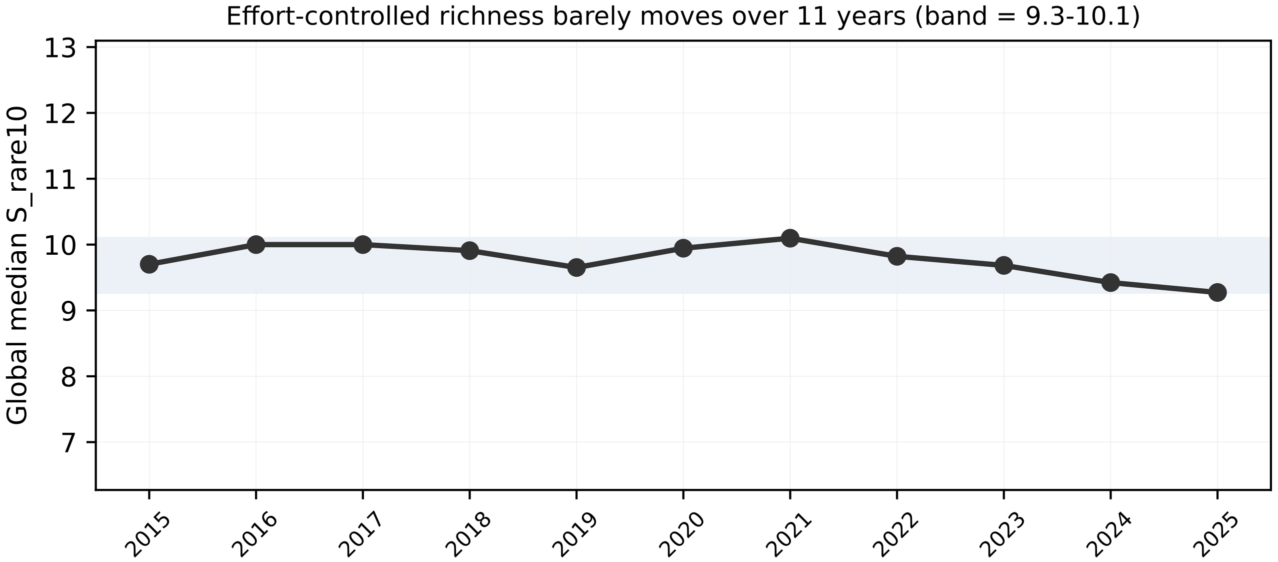
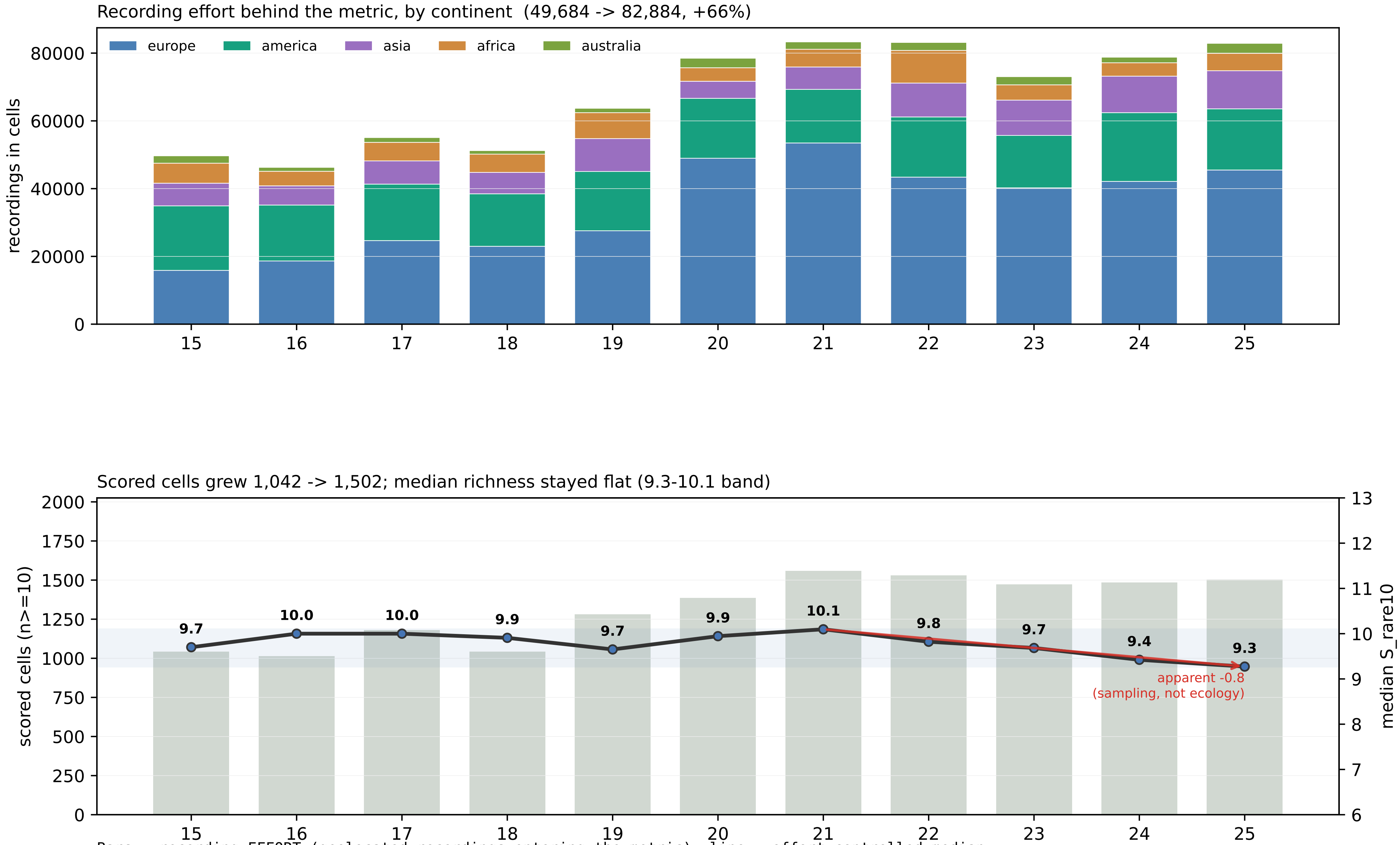


Year-by-year: effort vs. the metric (2015-2025)



Bars = recording EFFORT (geolocated recordings entering the metric); line = effort-controlled median richness. Effort rose ~70% over the decade while the metric stayed inside a narrow band. The apparent recent dip (red arrow) is NOT ecological degradation: it tracks the lower completeness of recent uploads and shifting recordist coverage. Do not read the decline as biodiversity loss.